CHE 201 Fall 2019 HW 7

Kathleen Sweeney

TOTAL POINTS

92 / 100

QUESTION 1

1 1 **15 / 15**

✓ **- 0 pts** Correct

- 1 pts a) Wrong schematic (A2 should be < A1)

- 1 pts a) wrong or missing system boundary

- 1.5 pts c) Missing mass balance

- 4 pts d) Wrong value, wrong calculation or missing T2. Correct answer is h2 = 99.5 Btu/lb and T2 = -2.4 oF

- 2 pts e) State 1 is superheated refrigerant

- 2 pts e) State 2 is saturated refrigerant (L+V mixture)

- 4 pts e) Wrong or missing P-v diagram

QUESTION 2

2 2 **11 / 15**

- 0 pts Correct

- 1 pts a) Wrong or missing schematic

- 1 pts a) wrong or missing system boundary

- 2 pts b) Wrong or missing eng model

- 1.5 pts c) Missing mass balance

- 2 pts d) Missing or wrong exit area. A2 = 0.06 m2

- 4 pts d) mass flow rate = 4.5 kg and A2 = 0.06 m2

- 2 pts e) V2 > V1 (expansion)

✓ **- 4 pts a) Missing or wrong T-v diagram**

QUESTION 3

3 3 **13 / 15**

- 0 pts Correct

- 1 pts a) Wrong schematic

- 1 pts a) Missing or wrong system boundary

- 2 pts b) State 1 is water vapor

- 4 pts b) Wrong or missing P-v diagram

✓ **- 2 pts b) V2 > V1 (expansion)**

- 2 pts b) State 2 is saturated vapor

- 2 pts c) Missing eng model

- 3 pts d) Missing simplified mass rate and energy rate balance.

- 1.5 pts d) Missing simplified mass rate

- 4 pts e) W = 3581 kW

QUESTION 4

4 4 **20 / 20**

✓ **- 0 pts** Correct

- 3 pts T-V figure

- 1 pts T-V figure, v1 = ? v2= ?

- 5 pts Q calculation incorrect

- 3 pts scheme incorrect, you mention dealta PE= 0 but your in put ant output is not on the same level

- 1 pts T-V figure where is step 1 and 2

- 3 pts scheme incorrect. input? output?

- 5 pts m incorrect

- 5 pts scheme

QUESTION 5

5 5 **15 / 15**

✓ **- 0 pts** Correct

- 3 pts scheme, storage tank that is 15 m above reservoir

- 3 pts storage tank that is 15 m above reservoir; the pump and the storage tank should be on the same level

- 5 pts w calculation incorrect

- 3 pts unit in KW

- 3 pts h1 = ? h2 = ?

- 3 pts engineering model

QUESTION 6

6 6 **18 / 20**

- 0 pts Correct

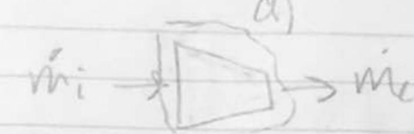
- 1 pts a) did not label the system boundary

- **- 1 pts** a) wrong schematic
- **- 4 pts** b)missing the engineering model
- **- 2 pts** c) missing the mass rate balance
- ✓ **- 2 pts d) missing or wrong energy rate balance**
- **- 4 pts** d) used wrong enthalpy values to solve for x
- **- 5 pts** d) x = 0.194
- **- 2.5 pts** e) State 1 is compressed liquid
- **- 5 pts** e) wrong or missing sketch of the process on a T-v diagram
- **- 2.5 pts** e) v2>v1
- **- 2.5 pts** e) missing point 2 on the T-v diagram
- **- 2.5 pts** e) T1>T2

HW 7

1) R-134A

a)



$T_i = 220°F$ $\quad V_e = 1500\ ft/s$

$V_i = 120\ ft/s$

$P_i = 200\ lbf/in^2$ $\quad P_e = \underline{20}\ lbf/in^2$

b) Model

- $\dot{Q}_{cv} = 0$ - $\dot{W}_{cv}$
- steady state
- ΔPE
- control volume
- one inlet one outlet

$h_1 = 144.15\ Btu/lb$

c) $0 = \dot{m}_i - \dot{m}_e \quad \dot{m}_i = \dot{m}_e = \boxed{\dot{m}}$

$$\frac{dE_{cv}}{dt} = 0 = \cancel{\dot{Q}_{cv}} - \cancel{\dot{W}_{cv}} + \dot{m}\left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g\cancel{(z_1 - z_2)}\right]$$

$$0 = (h_1 - h_2) + \frac{V_1^2 - V_2^2}{2}$$

d) find $T_e = ?$

$$h_2 = h_1 + \frac{V_1^2 - V_2^2}{2}$$

$$h_2 = 144.15\ Btu/lb + \left(\frac{120^2 - 1500^2}{2}\right)\frac{ft^2}{s^2}$$

$1\ Btu = ft \cdot lbf$

In T-v table
$h_f < h_2 < h_g$

ft^2	1 Btu	1 lbf
s^2	778.17 ft·lbf	32.174 lb·ft/s²

Look at table

$h_2 = 144.15\ Btu/lb - 44.6462\ Btu/lb$

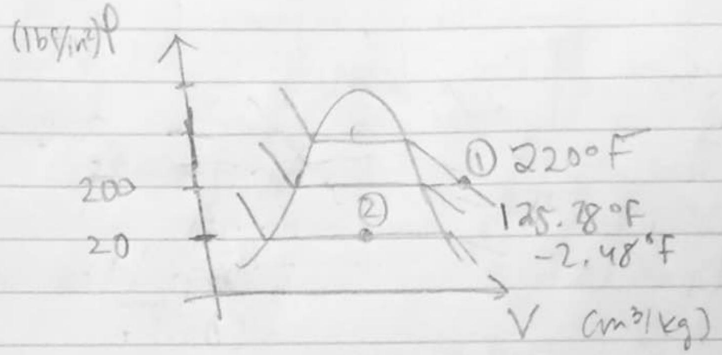
$h_2 = 99.50\ Btu/lb$

This is a liquid vapor mixture.

At $P_2 = 20\ lbf/in^2$

$\boxed{T = -2.48°F}$

e.

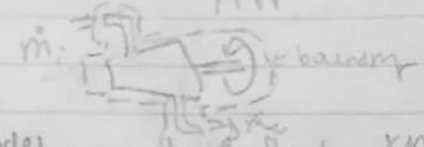


1 1 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 1 pts** a) Wrong schematic (A2 should be < A1)
- **- 1 pts** a) wrong or missing system boundary
- **- 1.5 pts** c) Missing mass balance
- **- 4 pts** d) Wrong value, wrong calculation or missing T2. Correct answer is h2 = 99.5 Btu/lb and T2 = -2.4 oF
- **- 2 pts** e) State 1 is superheated refrigerant
- **- 2 pts** e) State 2 is saturated refrigerant (L+V mixture)
- **- 4 pts** e) Wrong or missing P-v diagram

Air

2.



a)

b) Model

- steadystate
- control volume
- $\dot{W}_{cv} = 2500\ kW$
- $\cancel{\dot{Q}_{cv}}$
- $\cancel{\Delta PE}$
- ideal gas $pv = RT$
- one inlet one outlet

State known → State 2

$P_i = 8\ bar$ → $P_e = 1\ bar$

$T_i = 960K$ → $T_e = 450K$

$V_i = 0$ → $V_e = 90\ m/s$

$h_1 = 1000.55\ kJ/kg$ $h_2 = 451.80\ kJ/kg$

A-22 table A-22 $v = 223.6$

c) $\frac{dM_{cv}}{dt} = \dot{m}_i - \dot{m}_e = 0$ $\dot{m}_i = \dot{m}_e = \dot{m}$

$$\cancel{\frac{dE_{cv}}{dt}}^{0} = \cancel{\dot{Q}_{cv}}^{0} - \dot{W}_{cv} + \dot{m}\left[(h_1 - h_2) + \frac{\cancel{V_1^2}^{0} - V_2^2}{2} + g\cancel{(z_1 - z_2)}^{0}\right]$$

$$\boxed{\dot{W}_{cv} = \dot{m}\left[(h_1 - h_2) + \frac{-V_2^2}{2}\right]}$$

d) $\dot{m} = ?\ kg/s$

$A_e = ?$

$2{,}500\ kW =$

$2{,}500\ kJ/s = \dot{m}\left[(1000.55 - 451.80) + \left(\frac{0^2 - 90^2}{2}\right)\right]$ kJ/kg

$\frac{m^2}{s^2} \cdot \frac{1J}{1N \cdot m} \cdot \frac{1N}{1kg\ m/s^2} \cdot \frac{1kJ}{1000J} = kJ/kg$

$\cancel{kJ}/s \cdot \frac{kg}{\cancel{kJ}} = kg/s$

$$\boxed{\dot{m} = 4.59\ kg/s}$$

exit Area $\dot{m} = \frac{AV}{v}$

$4.59\ kg/s = \frac{A(90\ m/s)}{\left(\frac{0.287 \cdot 450K}{1\ bar}\right)}$

$\frac{kJ}{\cancel{K} \cdot kg} \cdot K \cdot \frac{N/m^2}{} \cdot \frac{1}{atm}$

$pv = RT$ $M_{air} = 28.97\ kg/kmol$

$v = \frac{RT}{p}$

$1\ bar \cdot \frac{10^5\ N/m^2}{1\ bar} \cdot \frac{1J}{1N \cdot m} \cdot \frac{1kJ}{1000J}$

$R = \frac{8.314\ kJ/kmol \cdot K}{28.97\ kg/kmol} = 0.2869884\ kJ/K \cdot kg$

$100\ \frac{kJ}{kg} \cdot \frac{m^3}{kJ} = \frac{m^3}{kg}$

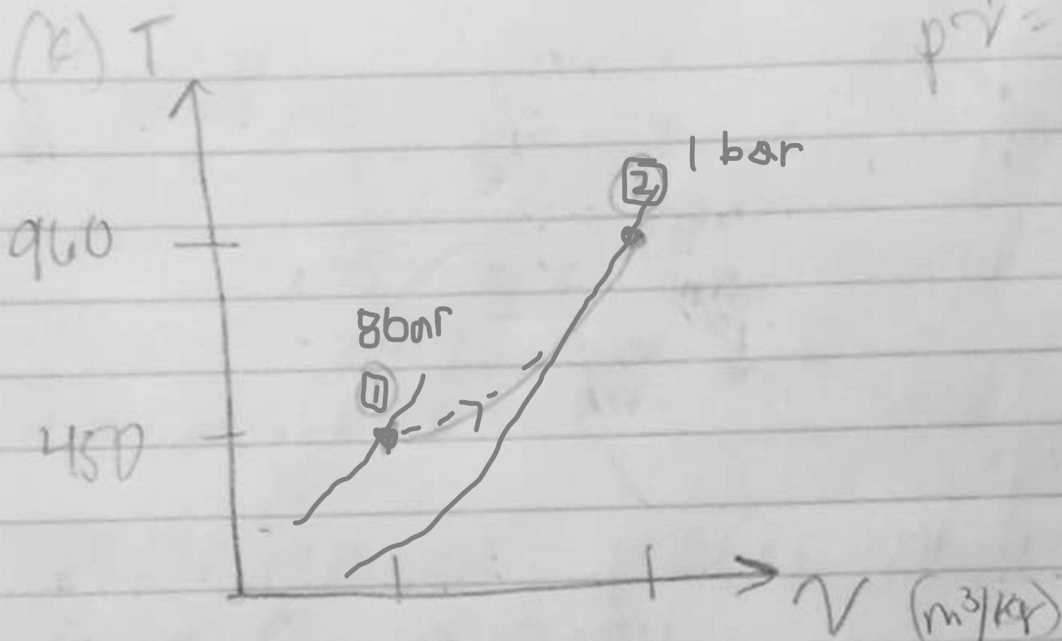
$4.59\ kg/s = \frac{A(90\ m/s)}{(1.2915\ m^3/kg)}$

$5.92799\ m^3/s = A(90\ m/s)$

$$\boxed{A_e = 0.066\ m^2}$$

$$\boxed{A_e = 0.066\ m^2}$$

e)

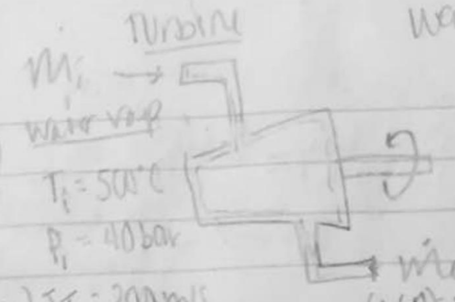


$p v = R T$

2 2 **11 / 15**

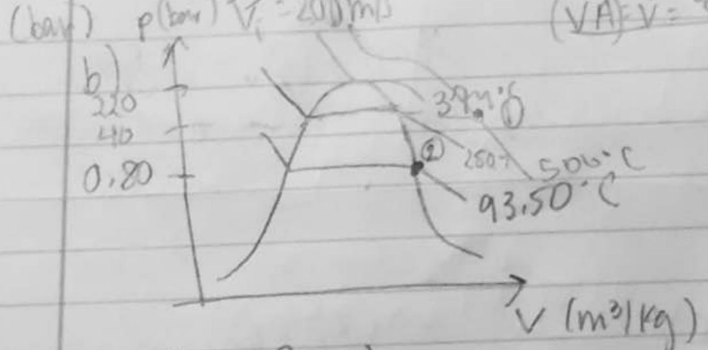
- **- 0 pts** Correct
- **- 1 pts** a) Wrong or missing schematic
- **- 1 pts** a) wrong or missing system boundary
- **- 2 pts** b) Wrong or missing eng model
- **- 1.5 pts** c) Missing mass balance
- **- 2 pts** d) Missing or wrong exit area. A2 = 0.06 m2
- **- 4 pts** d) mass flow rate = 4.5 kg and A2 = 0.06 m2
- **- 2 pts** e) V2 > V1 (expansion)
- ✓ **- 4 pts** **a) Missing or wrong T-v diagram**

3. a)



c) model:
- adiabatic (T is constant)
- Δpe
- Steady state
- one inlet one outlet

b)



d) $\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e = 0 = \dot{m}_i = \dot{m}_e = \boxed{\dot{m}}$

$$\frac{dE_{cv}}{dt} = 0 = \cancel{\dot{Q}_{cv}} - \dot{W}_{cv} + \dot{m}\left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + \cancel{g(z_1-z_2)}\right]$$

$$\boxed{\dot{W}_{cv} = \dot{m}\left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2}\right]}$$

e) $\dot{W}_{cv} = ?\ kW$

$\dot{m}_i = \dot{m}_e = \dot{m}$

$\frac{\cancel{m^3}}{s} \cdot \frac{kg}{\cancel{m^3}}$

table A-3 $\longrightarrow$ $\dot{m} = \frac{AV}{v} = \frac{9.48\ m^3/s}{2.087\ m^3/kg} = \boxed{4.54\ kg/s}$

T A-4

$h_1 = 3445.3$ kJ/kg

$h_2 = 2665.8$ kJ/kg

$$\dot{W}_{cv} = 4.54\ kg/s\left[(3445.3 - 2665.8)\ kJ/kg + \frac{\frac{200^2 - 150^2}{2}\left[\frac{m^2}{s^2}\right]}{1000}\right]$$

$\dot{W}_{cv} = 4.54$ kg/s $(788.25$ kJ/kg$)$

$3578.66\ \frac{kg}{s} \cdot \frac{kJ}{\cancel{kg}}$

$\dot{W}_{cv} = 3580\ kJ/s \left|\frac{1\ kW}{1\ kJ/s}\right. =$

$$\boxed{\dot{W}_{cv} = 3580\ kW}$$

3 3 **13 / 15**

- **- 0 pts** Correct
- **- 1 pts** a) Wrong schematic
- **- 1 pts** a) Missing or wrong system boundary
- **- 2 pts** b) State 1 is water vapor
- **- 4 pts** b) Wrong or missing P-v diagram
- ✓ **- 2 pts b) V2 > V1 (expansion)**
- **- 2 pts** b) State 2 is saturated vapor
- **- 2 pts** c) Missing eng model
- **- 3 pts** d) Missing simplified mass rate and energy rate balance.
- **- 1.5 pts** d) Missing simplified mass rate
- **- 4 pts** e) W = 3581 kW

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}[(h_1 - h_2)]$$

$$\dot{m}$$

$$0 = \frac{\dot{Q}_{cv}}{\dot{m}} - \frac{\dot{W}_{cv}}{\dot{m}} + [h_1 - h_2] \quad -\frac{\dot{Q}_{cv}}{\dot{m}} = -\frac{\dot{W}_{cv}}{\dot{m}} + (h_1 - h_2)$$

$$-\dot{Q}_{cv} = \left(\frac{\dot{W}_{cv}}{\dot{m}} + (h_1 - h_2)\right) \cdot \dot{m}$$

4. R-134A air compressor

a) $P_i = 4$ bar, $T_i = 20°C$, $(VA) = 4\ m^3/min$, $\dot{W}_{in} = -60\ kJ/kg$, $v_1 = 0.05397\ m^3/kg$

work on the system

$P_e = 12$ bar, $T_e = 80°C$

system boundary

$V\downarrow$

c) Model
- steady state
- Control volume
- $\cancel{\Delta KE}$
- $\cancel{\Delta PE}$
- One inlet one outlet

d)

$$\frac{dm_{cv}}{dt} = \dot{m}_{in} - \dot{m}_{out} = 0$$

$$\underline{\dot{m}_{in} = \dot{m}_{out} = \dot{m}}$$

$$\frac{dE_{cv}}{dt} = 0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}\left[(h_1 - h_2) + \cancel{\frac{V_1^2 - V_2^2}{2}} + \cancel{g(z_1 - z_2)}\right]$$

$$-\dot{Q}_{cv} = -\dot{W}_{cv} + \dot{m}[(h_1 - h_2)]$$

$$\boxed{\dot{Q}_{cv} = \dot{W}_{cv} - \dot{m}[h_1 - h_2]}$$

$\frac{m^3}{s} \cdot \frac{kg}{m^3}$

$$\dot{m} = \frac{AV}{v}$$

$$\dot{m} = \frac{4\ m^3/min\left(\frac{1}{60}\right)}{0.05397\ m^3/kg}$$

$$\underline{\dot{m} = 1.2352\ kg/s}$$

Table A-12

$h_1 = 262.96\ kJ/kg \quad h_2 = 310.24\ kJ/kg$

$$\frac{\dot{Q}_{cv}}{\dot{m}} = \dot{m}\left(\frac{W}{m} + (h_1 - h_2)\right)$$

$\frac{kg}{s} \cdot \frac{kJ}{kg}$

$$Q_{cv} = (-60\ kJ/kg) - 1.23525\ kg/s\ [-47.28\ kJ/kg]$$

$$\dot{Q}_{cv} = (1.23525\ kg/s)(-60\ kJ/kg) + 58.4026\ kJ/s$$

$$-74.115\ kJ/s + 58.4026\ kJ/s$$

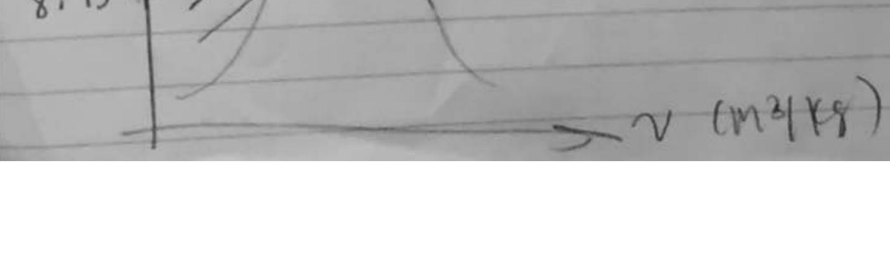
$\frac{kg}{s} \cdot \frac{kJ}{kg}$

$$\dot{Q}_{cv} = -15.71\ kJ/s =$$

$1\ kJ/s = 1\ kW$

$$\boxed{Q_{cv} = -15.71\ kW}$$

b)



4 4 **20 / 20**

- ✓ **- 0 pts** Correct
- **- 3 pts** T-V figure
- **- 1 pts** T-V figure, v1 = ? v2= ?
- **- 5 pts** Q calculation incorrect
- **- 3 pts** scheme incorrect, you mention dealta PE= 0 but your in put ant output is not on the same level
- **- 1 pts** T-V figure where is step 1 and 2
- **- 3 pts** scheme incorrect. input? output?
- **- 5 pts** m incorrect
- **- 5 pts** scheme

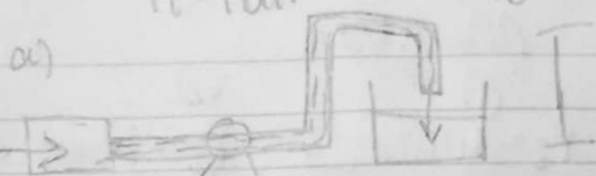
known:

$z_1 = 0m$, $T_i = 15°C$, $P_i = 1bar$ → $z_2 = 15m$, $T_f = 15°C$, $P_e = 3bar$ $\dot{m} = 1.5 kg/s$

$\boxed{h = u + PV}$

5. a)

liquid



$\Delta Z = 15 m$

$\dot{m} = 1.5 kg/s$ $V_1 = V_2$

$T_1 = T_2$

b) model
- steady state $\frac{dE_{cv}}{dt} = 0$ $\dot{m}_i = \dot{m}_e$
- control volume
- adiabatic
- $\cancel{\Delta KE}$ $-\Delta h = 0$
- $\cancel{\dot{Q}_{cv}}$ Δh is negligible

c) $\frac{dm_{cv}}{dt} = 0 = \dot{m}_i - \dot{m}_e$ $\boxed{\dot{m}_i = \dot{m}_e = \dot{m}}$

$$\frac{dE_{cv}}{dt} = 0 = \cancel{\dot{Q}_{cv}} - \dot{W}_{cv} + \dot{m}\left[(h_1 - h_2) + \cancel{\frac{V_1^2 - V_2^2}{2}} + g(z_1 - z_2)\right]$$

$h_1 = 62.99 kJ/kg + 1bar(1.0009\times10^{-3} m^3/kg)$

$h_2 = 62.99 kJ/kg + 3bar(1.0009\times10^{-3} m^3/kg)$

$h_2 = 63.2903 kJ/kg$

$$\boxed{\dot{W}_{cv} = \dot{m}\left[(h_1 - h_2) + g(z_1 - z_2)\right]}$$

$\frac{m^2}{s^2} \left| \frac{1J}{1N\cdot m} \right| \frac{1N}{1kg\cdot m/s^2} \left| \frac{1kJ}{1000J} \right.$

d) $\dot{W}_{cv} = ?\ kW$ $g = 9.81 m/s^2$

$$\dot{W}_{cv} = 1.5 kg/s\left[\left(\overset{h_1}{63.0901 \tfrac{kJ}{kg}} - \overset{h_2}{63.2903 kJ/kg}\right) + \frac{9.81 m/s^2 (0 - 15 m)}{1000}\right]$$

$$\frac{\cancel{kg}}{s} \cdot \frac{kJ}{\cancel{kg}} = \frac{kJ}{s} = 1kW$$

$\frac{m^3}{kg} \cdot \frac{N}{m^2}$ $N\cdot m$

$$\boxed{\dot{W}_{cv} = -0.52 kW}$$

$\frac{bar\cdot m^3/kg}{} \left| \frac{10^5 N/m^2}{1bar} \right| \frac{1J}{1N\cdot m} \left| \frac{1kJ}{1000J} \right. = 0.30027 kJ/kg$

5 5 **15 / 15**

✓ **- 0 pts** **Correct**

- 3 pts scheme, storage tank that is 15 m above reservoir

- 3 pts storage tank that is 15 m above reservoir; the pump and the storage tank should be on the same level

- 5 pts w calculation incorrect

- 3 pts unit in KW

- 3 pts h1 = ? h2 = ?

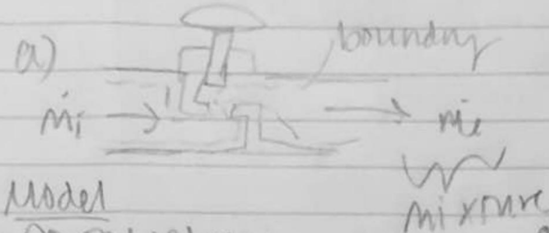
- 3 pts engineering model

$\Delta h = 0 \quad h_1 = h_2$

Ammonia

6.

a)



① $P_i = 10$ bar, $T_i = 24°C$ → ② $P_e = 1$ bar, mixture, $x = ?$

b) Model
- Control volume
- $\cancel{W_{cv}}$
- $\cancel{Q_{cv}}$
- $\cancel{\Delta PE}$
- throttling process $h_1 = h_2$
- $\cancel{\Delta h}$

c) $\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e = 0$

$\boxed{\dot{m}_i = \dot{m}_e = \dot{m}}$

$$\frac{dE_{cv}}{dt} = 0 = \cancel{\dot{Q}_{cv}} - \cancel{\dot{W}_{cv}} + \dot{m}\left[\cancel{(h_1 - h_2)} + \frac{V_1^2 - V_2^2}{2} + \cancel{g(z_1 - z_2)}\right]$$

$$\frac{0}{\dot{m}} = \frac{\dot{m}}{\dot{m}}\left(\frac{V_1^2 - V_2^2}{2}\right)$$

$V_1^2 - V_2^2 = 0 \quad \boxed{V_1 = V_2}$

d) $x = ?$

State 1 liquid

$h_1(T) \approx h_f(T)$

$h_1 = 293.4$ kJ/kg

$h_1 = h_2$

$h_2 = h_f + x(h_g - h_f)$

$h_f = 28.18$ kJ/kg

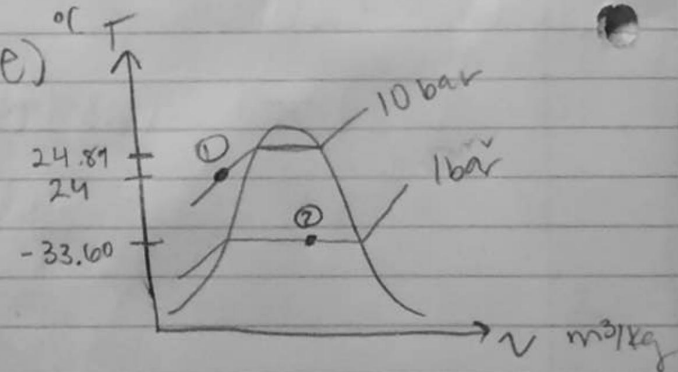
$h_g = 1398.4$ kJ/kg

$293.4 = 28.18 + x(1398.4 - 28.18)$

$x = 0.19356 \approx 0.194$

$\boxed{x = 0.194}$

e)



6 6 **18 / 20**

- **- 0 pts** Correct
- **- 1 pts** a) did not label the system boundary
- **- 1 pts** a) wrong schematic
- **- 4 pts** b)missing the engineering model
- **- 2 pts** c) missing the mass rate balance
- ✓ **- 2 pts d) missing or wrong energy rate balance**
- **- 4 pts** d) used wrong enthalpy values to solve for x
- **- 5 pts** d) x = 0.194
- **- 2.5 pts** e) State 1 is compressed liquid
- **- 5 pts** e) wrong or missing sketch of the process on a T-v diagram
- **- 2.5 pts** e) v2>v1
- **- 2.5 pts** e) missing point 2 on the T-v diagram
- **- 2.5 pts** e) T1>T2